Practical No. 6

PRN: 22510019

Name: Soham Patil

Batch: B6

Title of practical:

Installation of MPI & Implementation of basic functions of MPI

```
This message is shown once a day. To disable it please create the
/home/soham/.hushlogin file.
soham@Vivobook16x-Soham:/mnt/c/Users/asus$
soham@Vivobook16x-Soham:/mnt/c/Users/asus$ sudo apt install -y mpich
[sudo] password for soham:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  gfortran gfortran-13 gfortran-13-x86-64-linux-gnu gfortran-x86-64-linux-gnu hwloc-nox ibverbs-providers
  libamd-comgr2 libamdhip64-5 libevent-pthreads-2.1-7t64 libgfortran-13-dev libgfortran5 libhsa-runtime64-1 libhsakmt1
  libhwloc-plugins libhwloc15 libibverbs1 libmpich-dev libmpich12 libmunge2 libnl-3-200 libnl-route-3-200 libnuma1
  libpmix2t64 librdmacm1t64 libslurm40t64 libucx0 libxnvctrl0 ocl-icd-libopencl1
Suggested packages:
  gfortran-multilib gfortran-doc gfortran-13-multilib gfortran-13-doc libcoarrays-dev libhwloc-contrib-plugins
  mpich-doc opencl-icd
The following NEW packages will be installed:
  gfortran gfortran-13 gfortran-13-x86-64-linux-gnu gfortran-x86-64-linux-gnu hwloc-nox ibverbs-providers
  libamd-comgr2 libamdhip64-5 libevent-pthreads-2.1-7t64 libgfortran-13-dev libgfortran5 libhsa-runtime64-1 libhsakmt1
  libhwloc-plugins libhwloc15 libibverbs1 libmpich-dev libmpich12 libmunge2 libnl-3-200 libnl-route-3-200 libnuma1
  libpmix2t64 librdmacm1t64 libslurm40t64 libucx0 libxnvctrl0 mpich ocl-icd-libopencl1
0 upgraded, 29 newly installed, 0 to remove and 101 not upgraded.
Need to get 60.2 MB of archives.
After this operation, 245 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libnl-3-200 amd64 3.7.0-0.3build1.1 [55.7 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libnl-route-3-200 amd64 3.7.0-0.3build1.1 [189 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libibverbs1 amd64 50.0-2ubuntu0.2 [68.0 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 ibverbs-providers amd64 50.0-2ubuntu0.2 [381 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble/main amd64 libnuma1 amd64 2.0.18-1build1 [23.3 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libgfortran5 amd64 14.2.0-4ubuntu2~24.04 [916 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libgfortran-13-dev amd64 13.3.0-6ubuntu2~24.04 [928 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 gfortran-13-x86-64-linux-gnu amd64 13.3.0-6ubuntu2~24.04 [11.4 MB]
Get:9 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 gfortran-13 amd64 13.3.0-6ubuntu2~24.04 [13.9 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble/main amd64 gfortran-x86-64-linux-gnu amd64 4:13.2.0-7ubuntu1 [1024 B]
```

Verification :

```
soham@Vivobook16x-Soham:/mnt/c/Users/asus$ mpicc --version
gcc (Ubuntu 13.3.0-6ubuntu2~24.04) 13.3.0
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This is free software; see the source for copying conditions.  There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

soham@Vivobook16x-Soham:/mnt/c/Users/asus$ mpirun --version
HYDRA build details:
    Version:                                 4.2.0
    Release Date:                            Fri Feb  9 12:29:21 CST 2024
    CC:                              gcc -Wdate-time -D_FORTIFY_SOURCE=3 -g -O2 -fno-omit-frame-pointer -mno-omit-leaf-frame-pointer -ffile-prefix-map=/buil
d/mpich-C3oteG/mpich-4.2.0=. -flto=auto -ffat-lto-objects -fstack-protector-strong -fstack-clash-protection -Wformat -Werror=format-security -fcf-protection
 -fdebug-prefix-map=/build/mpich-C3oteG/mpich-4.2.0=/usr/src/mpich-4.2.0-5build3 -Wl,-Bsymbolic-functions -flto=auto -ffat-lto-objects -Wl,-z,relro
    Configure options:                       '--with-hwloc-prefix=/usr' '--with-device=ch4:ucx' 'FFLAGS=-O2 -fno-omit-frame-pointer -mno-omit-leaf-frame-poi
nter -ffile-prefix-map=/build/mpich-C3oteG/mpich-4.2.0=. -flto=auto -ffat-lto-objects -fstack-protector-strong -fstack-clash-protection -fcf-protection -fde
bug-prefix-map=/build/mpich-C3oteG/mpich-4.2.0=/usr/src/mpich-4.2.0-5build3 -fallow-invalid-boz -fallow-argument-mismatch' '--prefix=/usr' 'CFLAGS=-g -O2 -f
no-omit-frame-pointer -mno-omit-leaf-frame-pointer -ffile-prefix-map=/build/mpich-C3oteG/mpich-4.2.0=. -flto=auto -ffat-lto-objects -fstack-protector-strong
 -fstack-clash-protection -Wformat -Werror=format-security -fcf-protection -fdebug-prefix-map=/build/mpich-C3oteG/mpich-4.2.0=/usr/src/mpich-4.2.0-5build3'
'LDFLAGS=-Wl,-Bsymbolic-functions -flto=auto -ffat-lto-objects -Wl,-z,relro' 'CPPFLAGS=-Wdate-time -D_FORTIFY_SOURCE=3'
    Process Manager:                         pmi
    Launchers available:                     ssh rsh fork slurm ll lsf sge manual persist
    Topology libraries available:            hwloc
    Resource management kernels available:   user slurm ll lsf sge pbs cobalt
    Demux engines available:                 poll select
soham@Vivobook16x-Soham:/mnt/c/Users/asus$
```

Problem Statement 1:

Implement a simple hello world program by setting number of processes equal to 10

Program

Assignment6 > ps1.c > ...

```c
#include <mpi.h>
#include <stdio.h>

int main(int argc, char** argv) {
    MPI_Init(&argc, &argv);

    int world_rank, world_size;

    MPI_Comm_rank(MPI_COMM_WORLD, &world_rank);
    MPI_Comm_size(MPI_COMM_WORLD, &world_size);

    printf("Hello from process %d out of %d processes\n", world_rank, world_size);

    MPI_Finalize();
    return 0;
}

```

```
soham@Vivobook16x-Soham:/mnt/d/SEM VII/HPCL/Assignment6$ mpicc ps1.c -o ps1
soham@Vivobook16x-Soham:/mnt/d/SEM VII/HPCL/Assignment6$ mpirun --oversubscribe -np 10 ./ps1
Hello from process 0 out of 10 processes
Hello from process 5 out of 10 processes
Hello from process 6 out of 10 processes
Hello from process 7 out of 10 processes
Hello from process 8 out of 10 processes
Hello from process 9 out of 10 processes
Hello from process 1 out of 10 processes
Hello from process 2 out of 10 processes
Hello from process 3 out of 10 processes
Hello from process 4 out of 10 processes
soham@Vivobook16x-Soham:/mnt/d/SEM VII/HPCL/Assignment6$ |
```

Information :

- MPI_Init is used to start the MPI environment, and it must be the first MPI call.
- MPI_Comm_rank gives the rank (ID) of the current process. Each process gets a unique number.
- MPI_Comm_size gives the total number of processes running in the communicator.
- Each process prints a message with its rank and the total number of processes.
- MPI_Finalize is called at the end to cleanly stop the MPI environment.
- The output shows 10 different “Hello” messages, each from a different process (ranks 0–9), which means the parallel execution worked correctly.

Problem Statement 2:

Implement a program to display rank and communicator group of five processes

Program:

Assignment6 > ps2.c > main(int, char **)

```c
#include <mpi.h>
#include <stdio.h>

int main(int argc, char** argv) {
    MPI_Init(&argc, &argv);

    int world_rank, world_size;
    MPI_Comm_rank(MPI_COMM_WORLD, &world_rank);
    MPI_Comm_size(MPI_COMM_WORLD, &world_size);

    printf("Process %d out of %d in MPI_COMM_WORLD\n", world_rank, world_size);
    fflush(stdout);

    MPI_Barrier(MPI_COMM_WORLD);

    MPI_Comm new_comm;
    MPI_Comm_split(MPI_COMM_WORLD, 0, world_rank, &new_comm);

    int new_rank, new_size;
    MPI_Comm_rank(new_comm, &new_rank);
    MPI_Comm_size(new_comm, &new_size);

    printf("Process %d in new communicator group of size %d\n", new_rank, new_size);
    fflush(stdout);

    MPI_Finalize();
    return 0;
}
```

Screenshot:

```
soham@Vivobook16x-Soham:/mnt/d/SEM VII/HPCL/Assignment6$ mpicc ps2.c -o ps2
soham@Vivobook16x-Soham:/mnt/d/SEM VII/HPCL/Assignment6$ mpirun --oversubscribe -np 7 ./ps2
Process 6 out of 7 in MPI_COMM_WORLD
Process 0 out of 7 in MPI_COMM_WORLD
Process 1 out of 7 in MPI_COMM_WORLD
Process 2 out of 7 in MPI_COMM_WORLD
Process 3 out of 7 in MPI_COMM_WORLD
Process 4 out of 7 in MPI_COMM_WORLD
Process 5 out of 7 in MPI_COMM_WORLD
Process 3 in new communicator group of size 7
Process 4 in new communicator group of size 7
Process 5 in new communicator group of size 7
Process 6 in new communicator group of size 7
Process 0 in new communicator group of size 7
Process 1 in new communicator group of size 7
Process 2 in new communicator group of size 7
soham@Vivobook16x-Soham:/mnt/d/SEM VII/HPCL/Assignment6$
```

Q3: Implement a MPI program to give an example of Deadlock.

Program:

```c
#include <mpi.h>
#include <stdio.h>

int main(int argc, char* argv[]) {
    int rank;
    MPI_Init(&argc, &argv);
    MPI_Comm_rank(MPI_COMM_WORLD, &rank);

    int data = rank;

    if (rank == 0) {
        // Deadlock: P0 waits for recv but P1 is also waiting
        MPI_Recv(&data, 1, MPI_INT, 1, 0, MPI_COMM_WORLD, MPI_STATUS_IGNORE);
        MPI_Send(&data, 1, MPI_INT, 1, 0, MPI_COMM_WORLD);
    } else if (rank == 1) {
        MPI_Recv(&data, 1, MPI_INT, 0, 0, MPI_COMM_WORLD, MPI_STATUS_IGNORE);
        MPI_Send(&data, 1, MPI_INT, 0, 0, MPI_COMM_WORLD);
    }

    MPI_Finalize();
    return 0;
}

```

Screenshot:

```
soham@Vivobook16x-Soham:/mnt/d/SEM VII/HPCL/Assignment6$ mpicc ps3.c -o ps3
soham@Vivobook16x-Soham:/mnt/d/SEM VII/HPCL/Assignment6$ mpirun --oversubscribe -np 5 ./ps3
```

Q4. Implement blocking MPI send & receive to demonstrate Nearest neighbor exchange of data in a ring topology.

Program

```c
D:\SEM VII\HPCL\Assignment6 • Contains emphasized items
#include <stdio.h>

int main(int argc, char* argv[]) {
    int rank;
    MPI_Init(&argc, &argv);
    MPI_Comm_rank(MPI_COMM_WORLD, &rank);

    int data = rank;

    if (rank == 0)
    {
        MPI_Recv(&data, 1, MPI_INT, 1, 0, MPI_COMM_WORLD, MPI_STATUS_IGNORE);
        MPI_Send(&data, 1, MPI_INT, 1, 0, MPI_COMM_WORLD);
    } else if (rank == 1) {
        MPI_Recv(&data, 1, MPI_INT, 0, 0, MPI_COMM_WORLD, MPI_STATUS_IGNORE);
        MPI_Send(&data, 1, MPI_INT, 0, 0, MPI_COMM_WORLD);
    }

    MPI_Finalize();
    return 0;
}
```

Screenshot:

```
Process 2 sent 2 to 3 and received 1 from 1
Process 1 sent 1 to 2 and received 0 from 0
Process 3 sent 3 to 4 and received 2 from 2
Process 4 sent 4 to 0 and received 3 from 3
Process 0 sent 0 to 1 and received 4 from 4
```

Q5. Write a MPI program to find the sum of all the elements of an array A of size

n. Elements of an array can be divided into two equals groups. The first [n/2]

elements are added by the first process, P0, and last [n/2] elements the by second process, P1. The two sums then are added to get the final result.

Program

```c
int main(int argc, char** argv) {
    MPI_Init(&argc, &argv);

    int rank;
    MPI_Comm_rank(MPI_COMM_WORLD, &rank);

    int n = 10;
    int A[10] = {7,9,6,-1,55};
    int partial_sum = 0;
    int total_sum = 0;

    int half = n / 2;

    if(rank == 0) {
        for(int i = 0; i < half; i++)
            partial_sum += A[i];

        printf("Process 0 partial sum: %d\n", partial_sum);
        MPI_Send(&partial_sum, 1, MPI_INT, 1, 0, MPI_COMM_WORLD);
    }
    else if(rank == 1) {

        for(int i = half; i < n; i++)
            partial_sum += A[i];
        printf("Process 1 partial sum: %d\n", partial_sum);

        int sum_from_p0;
        MPI_Recv(&sum_from_p0, 1, MPI_INT, 0, 0, MPI_COMM_WORLD, MPI_STATUS_IGNORE);
        total_sum = partial_sum + sum_from_p0;
        printf("Total sum of array elements: %d\n", total_sum);
    }

    MPI_Finalize();
    return 0;
}
```

Screenshot:

```
soham@Vivobook16x-Soham:/mnt/d/SEM VII/HPCL/Assignment6$ mpicc ps5.c -o ps5
soham@Vivobook16x-Soham:/mnt/d/SEM VII/HPCL/Assignment6$ mpirun --oversubscribe -np 2 ./ps5
Process 0 partial sum: 76
Process 1 partial sum: 0
Total sum of array elements: 76
soham@Vivobook16x-Soham:/mnt/d/SEM VII/HPCL/Assignment6$
```

Github Link : HPCL/Assignment6 at main · Somzee5/HPCL